

Network security

Unit 1 Introduction

Attack, Services and Mechanism,
Model for Internet security (8-1)

Unit 2 Cryptography

Cryptography: Notion of Plain Text,
Encryption, Key, Cipher Text,
Decryption and cryptanalysis:
Public Key Encryption (2)

Unit 3 Digital signature

Overview, Digital signature and
Authentication (2)

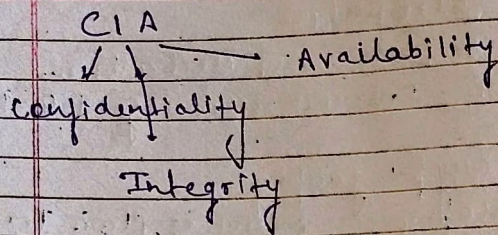
Unit 4 Web security (1)

Requirement, Secure Socket Layer,
Transport Layer Security,
and Secure Electronic Transactions

Unit 5 System security (2)

Introduction, Viruses and
Related Threats, Firewall, Design
Principles.

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Attack

Passive attack
(not modify the message, read)

Active Attack
(Modify the message)

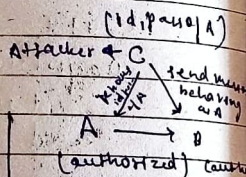
1) Masquerade

Release of message

Traffic analysis

encryption

location, address, of host, frequency



2) Modification of data

Denial of Service
(a network of attackers continuously load the server by asking services, causes servers to get down)

3) Non-repudiation

(refuse to accept that he/she sent message)

4) Replay

continuously sending the same message to B. B confused

Passive

- not detected
- focus prevention
- not modify the data

eg - release of message, Traffic analysis

Active

- detected
- focus and detection and correction
- modify the data

eg - Replay

Q1) what is attack?

Q2) what are different types of attack

Q3) Differentiate passive attack & active attack

Security services

- 1) Authentication - not every user allowed to use resources
- 2) Access control - make files & resources data read only

Authentication

- Peer to peer Authentication
- Data origin authentication

- 3) Data Confidentiality - only it is readable for destination, not for everyone
- 4) Data Integrity - No modification of data
- 5) Non-repudiation - neither the sender, nor the receiver can deny the exchange of message

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Security Mechanism

Implementation of security services is called security mechanism

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Network Security Mechanisms

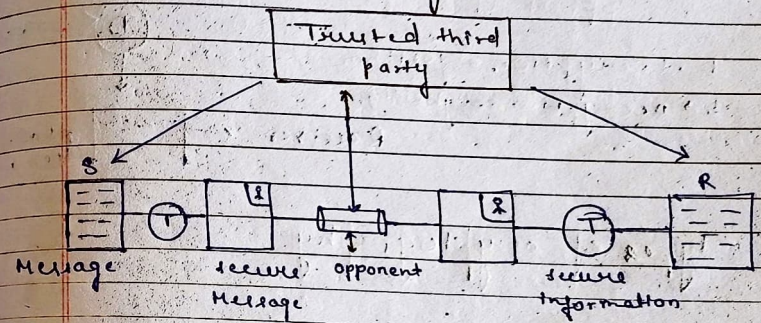
- 1) Encryption - To implement confidentiality, unreadable format
- 2) Digital signature - Implements Authentication and Non-repudiation
- 3) Access control - Gives access rights
- 4) Data integrity - Prevents modification of message
- 5) Data Authentication - Exchange Exchanging some piece of information periodically
- 6) Traffic Padding - Prevent traffic analysis

Date: / /

exchange/transmits dummy data streams

- 7) Routing Control - Leaves the suspected routes
- 8) Notarization - Trusted third party, SSL certificate - (https)

Network Security Model



S - sender

T - secure information transportation

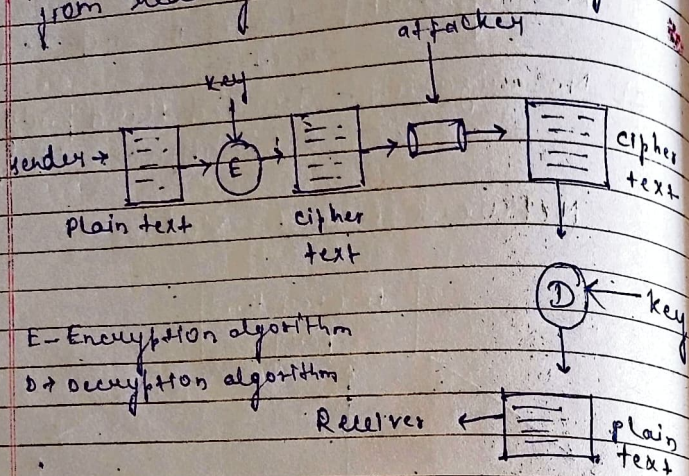
R - Receiver

Cryptography

Cryptography is the study of secure communication technique that allow only the sender and intended of a message to view its contents. Cryptography is about constructing and analyzing protocol that

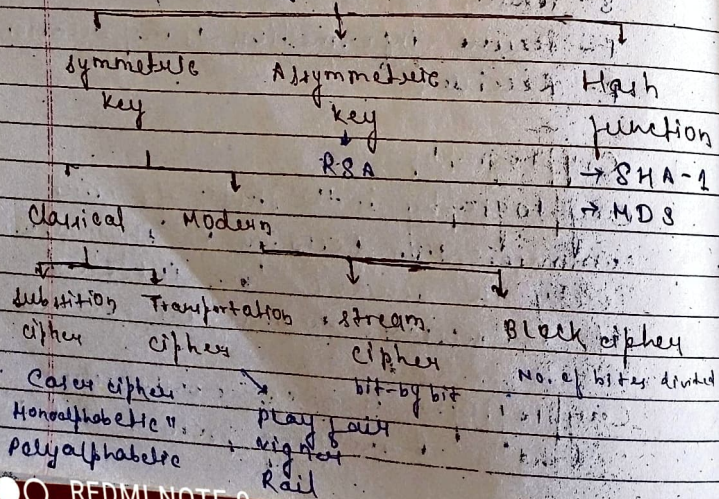
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prevent third party or the public
from reading private messages



Types of cryptography

cryptography

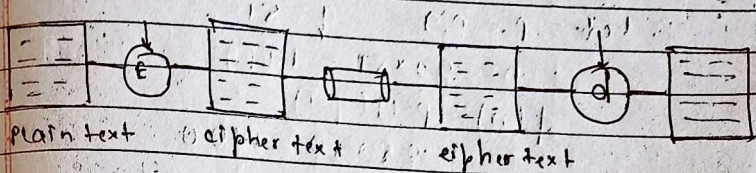


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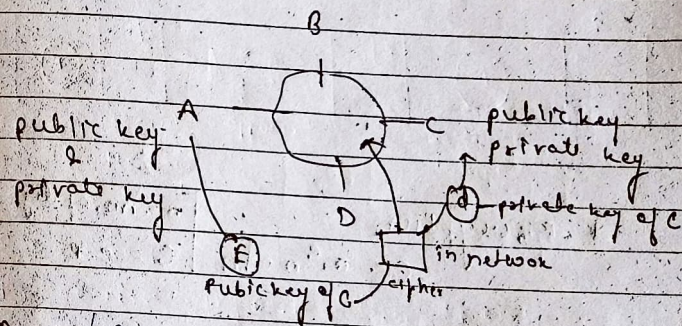
Hash function

SHA-1 → Secure Hashing algorithm
MD5 → Message digest

× Symmetric key



→ Asymmetric key / public key



RSA (Rivest, Shamir, Adleman)

- step 1: Take any two large numbers P and Q
 - 2: calculate $n = P * Q$
 - 3: calculate $\phi(n) = (P-1)(Q-1)$
 - 4: choose the value of e such that $1 < e < \phi(n)$ and $\text{gcd}(\phi(n), e) = 1$
- gcd - greatest common divisor

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5. calculate $d = e^{-1} \bmod \phi(n)$
 6. Public key = $\{e, n\}$
 7. Private key = $\{d, n\}$

$$\therefore \text{let } p = 29, q = 51$$

$$\therefore n = 29 \times 51 = 1479$$

$$\phi(n) = (p-1)(q-1)$$

$$= 28 \times 50 = 1400$$

$$e = 1399$$

* encryption

$$c = m^e \bmod n$$

$m = \text{no. of digits in PT}$
$e = 1399$

* decryption $m = c^d \bmod n$

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DES (Data Encryption Standard)
 Block cipher algorithm

has total 16 rounds

Text size = 64 bits $\Rightarrow$ Plaintext

Key size = 48 bits

16 bits removed

8 bits for parity

8 bits for rearrangement

$\rightarrow$ each round performs 4 steps

step 1: Dividing 64 bits in two parts, each 32 bits.

step 2: Bit shuffling

step 3: Non-linear substitution

step 4: Exclusive-OR operation

Initial key

64 bit

Permitted

choice 1

PCT

8 bits removed for parity bit

Initially from 64 bit key, each parity bit is removed from every 8th bit.

they apply left circular shift (LCS)

after dividing 56 bits into two parts

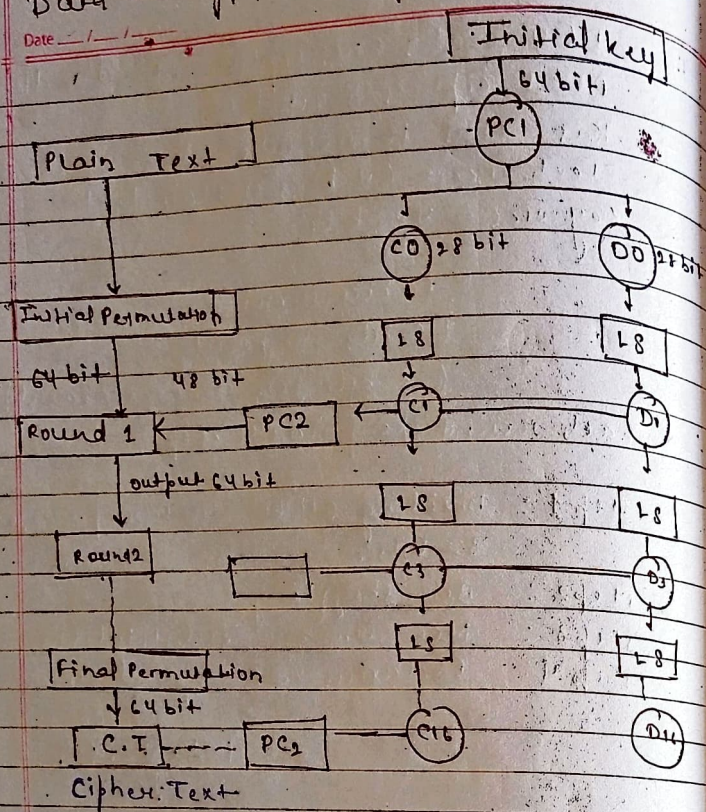
each having 28 bits

left circular shift - shift the bits based on round number

As a result we get C1 and D1

Data Encryption Algorithm

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Digital signature

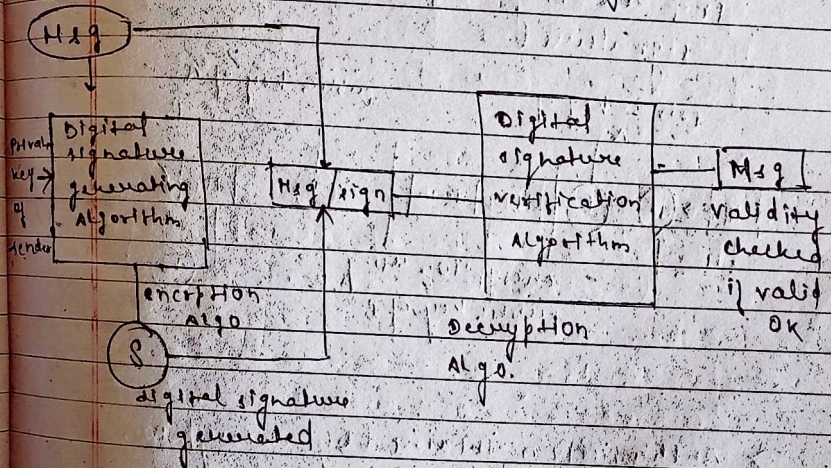
→ Integrity and Authentication
non-repudiation

A digital signature is a mathematical technique used to validate the authenticity and integrity of a digital document or message or software.

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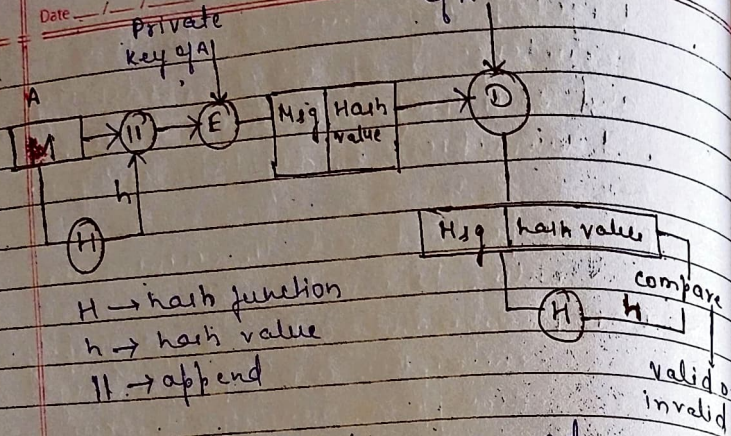
Digital signature can provide evidence of origin, identity and status of electronic documents, transaction or digital messages.

- play important role in e-commerce, online transaction etc.
- Based asymmetric key cryptography
 - encryption using private key
 - decryption using public key
- used for message authentication, integrity and non-repudiation
- not used for confidentiality



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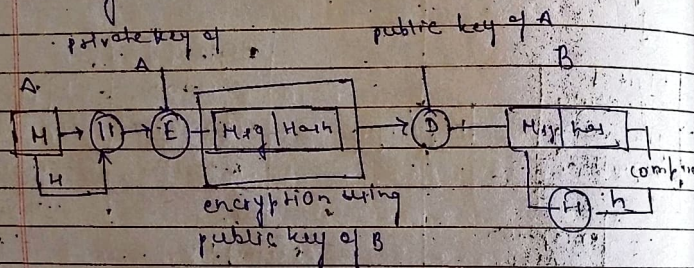
Public key
of A



For digital signature, we need

- 1) Key generation algorithm
- 2) signing algorithm
- 3) verification algorithm

For adding confidentiality in digital signature



- Authenticity maintained by encryption using private key of A
- Integrity by hash function
- Non-repudiation by public key and private key concept

Date: / / unit-5

Malware $\rightarrow$ Malicious software

Types-

- i. Virus
- ii. Trojan horse
- iii. Phishing
- iv. Ransomware
- v. Spyware
- vi. Botnet

Features of Malware

1. Lack of storage space
2. Programs open and close automatically
3. Emails and messages being sent automatically
4. Increased pop-ups, toolbars and other unwanted programs

Virus

- A kind of malicious computer program which when executed replicates itself and inserts its code.
- When the replication is done, this code infects the other files and programs present on your present computer system.
- It increases response time and devices will not work.
- Due to virus data will loss, programs will get corrupted and we will lose our control on the computer system.

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Trojan horse

- Looks like a desirable code or software.
- Take control of victim's system for malicious purposes, once downloaded.
- It is hidden piece of code which steals important information.
- Present in games, and apps.
- Embedded in attachments, included in phishing emails.

Worms

- Worms are like virus, they replicate but not change the program. In other words, it slows down the functionality of computer system.
- Spreads by a phishing attack.
- Modify and delete files.
- Install a convenient back door for hackers.

Phishing

Phishing is a type of social engineering attack, a common method of cyber attack.

- Spread through emails, text messages and web links.

Ransom ware

Uses encryption algorithm

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Introducer

They are attackers who attempt to breach the security of a network.

Types

1) Masquerader

They are unauthorized persons who use our system and exploit legitimate

Date: / /

User account They are usually outsiders.

2) Mistaken

legitimate users only and authorized but he/she will misuse their privileges. Usually they are insider.

3) Clandestine user

A person who takes the supreme control of the system and stops access to legitimate user.

Counter measures for all malware

1. Prevention ^{eg-} → Antivirus, hard to prevent
2. Detection - whether malware is present or not.
3. Identification - In which area of system (memory, id pass, deletion-)
4. Removal/Replacement of code or software or application

Intrusion Detection system (IDS)

- Detects the intrusion
- It continuously monitors the network and check for any malicious content is in network.
- Works in background.

Date: / /

If any intrusion is detected, IDS send "alert" to system admin to take necessary action.

Types of IDS

1. Network IDS (NIDS)
2. Host IDS (HIDS)

1. NIDS

- It monitors, analyses and captures the network traffic and detects the malicious data.
- It has predefined list of all possible attacks.
- Any match found, it identifies and notify to system admin that there is an attack.
- Difficult to implement in large network.

2. Host Intrusion Detection system

- Host-based (works on individual host or device)
- same perform, same process as NIDS

Identification

- Initially, it takes a snapshot of the clean system.
- Later, it takes snapshot of existing system and compares with clean.

system, so no change means no attack. And if there is a change, it means there is an attack after that it notifies the system admin.

Firewall

- Network security system that monitors and control incoming and outgoing traffic based on some predefined security rules.

Firewall Design principles

- Developing security policy - according to client requirement, according to type of traffic of client.

a) simple design

- A simple design is easier to maintain.
- we can make upgrades in the simple design according to the new possible threats.

b) selecting/choosing right device

- fully secured, well designed, not used outdated system.

c) layered defence

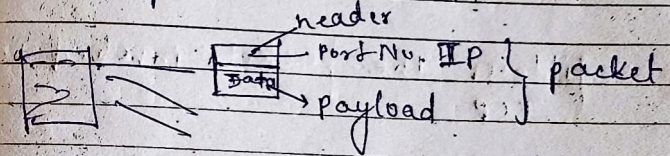
- has to be multilayered.

- consider internal threats.

Types of Firewall

1. Packet filtering firewall applies set of rules on each incoming packet and forwards the packet if rules are obeyed otherwise discard (drop/leave) the packets.

- Allowed or disallowed IP addresses
- checks packet type
- checks port number



These rules are based on source IP, Destination IP, protocols and ports. Also maintains a filtering.

2. Circuit level Firewall

Transport layer - error control, flow control, segmentation, reassembly, responsible for process delivery of message.

Data-link layer - error control, flow control, responsible for delivery of message.

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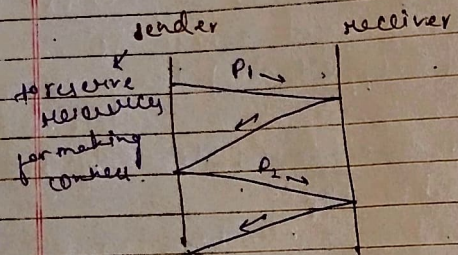
all layers have protocols except

TCP/IP
suite

Transport layer (TCP, UDP, SCTP)

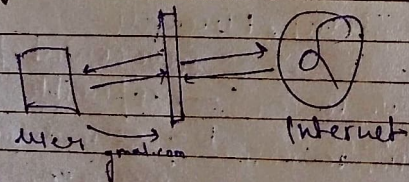
stream control transfer
connection oriented
connection less protocol

SMTP } supported by
FTP } Transport layer by TCP



filters the packets involved in the TCP

8. Application level gateway or (Proxy Firewall)



① It is more secure than packet
REDMI NOTE 9
firewall ② contacts user
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by TCP/IP protocols (or application layer protocols).

C. W
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Types of Firewall

1. Packet Filtering Firewall.

→ Applies set of rules on each incoming packets & forwards the packet if rules are obeyed otherwise discard (drop/leave) the packet.

Allowed or disallowed IP addresses
checks packet type, checks port number

These rules are based on source IP, destination IP, protocol & ports. Also maintains a Filtering.

2. Circuit level ^{gateway} Firewall

3. Application level gateway or Proxy Firewall.

→ It is more secure than packet Filtering Firewall.

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Type of Firewall

4. Hybrid Firewall :-

Web-security

Web security refers to networks & computer system & data are protected from unauthorized person or group.

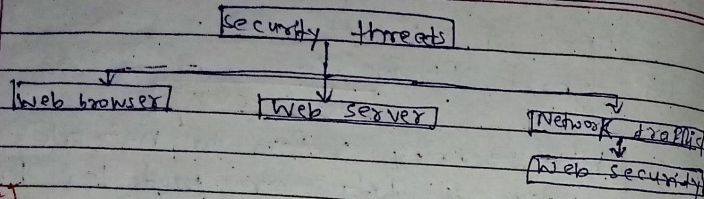
The purpose is to prevent security attack

- i) Modification of message.
- ii) Denial of service.
- iii) Phishing.
- iv) SQL injection
- v) Malware

How can achieve web-security

There are various tool & technology are available to achieve web security

1. Web & network Firewall
2. Keep your software update
3. Backup your data
4. Keep your website clean
5. Use strong password policy
6. Use password cracking tools.
7. Scanned your website for vulnerabilities
8. Use Anti-virus.

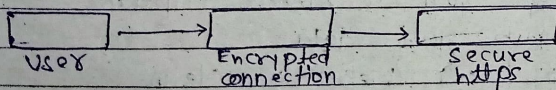


Secure Socket layer

Transport layer

SSL is developed by Netscape communication. It is protocol for establishing for secure links b/w network computer. SSL provides confidentiality, authentication & data integrity in internet communication.

An SSL certificate is bit of code on our web server that provide security for online connection.



* SSL Architecture

SSL works b/w Application layer & transport layer. It is a two layer

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IP Address } Socket
Port Address }

- i) change spec protocol :- Use of require cypher technique for data encryption
- ii) SSL alert protocol :- Alert (warning error if any) generation.
- iv) SSL record protocol :- Encrypted data transmission & encapsulation of the data send by the higher layer protocol

SSL Handshake	change spec	SSL Alert	HTTP
SSL record protocol			
Transport layer			
Network layer			
Datalink & Physical			

Two important term in SSL

- i) SSL connection :- It is a transport that provides a suitable types of service. Each connection is associated with one SSL session.
- ii) SSL session :- It is a set of cryptographic security parameter which can be shared among multiple SSL connection. An SSL session is a association b/w a client & server

* SSL Protocol

1. SSL handshake protocol :-

Phase 1

Establishing security capabilities

① Client hello $\Rightarrow$ The highest SSL version which the client can support.

- ② A session id that defines the session
- $\rightarrow$ There is a cipher suite parameter that contains the entire cryptographic algorithm which supports client system.
 - $\rightarrow$ A list of compression methods that can be supported by client system.

③ Server hello

- $\rightarrow$ The highest SSL which the server can support.
- $\rightarrow$ A session id that defines the session.
- $\rightarrow$ A cipher suite of contains the list of all cryptographic algorithms that is sent by client which the server can select the algorithm.
- $\rightarrow$ A list of compression methods sent by the client from which server will select the method.

Phase 2

server authentication & key exchange

① certificate

$\rightarrow$ The server sent the certificate message to authentication itself to the client.

② server key exchange

$\rightarrow$ This is optional. It is used only if the server doesn't send ~~the~~ digital certificate to client.

③ certificate request

$\rightarrow$ The server can request for the digital certificate to the client.

④ Server hello message

$\rightarrow$ The server ~~msg~~ hello done is the last message in phase 2, this indicates the client that the client can now verify all the certificate received by the server.

Phase 3

client authentication & key exchange

① client certificate

② client key exchange



→ The client send the client key exchange, the content in this are based on key exchange algorithm b/w both the parties.

(iii) certificate verify

Phase 4
Finished.

① change cypher spec

→ It is a client side messages telling about the current status of cypher protocol & parameter which has been made active from pending state.

(ii) Finished

→ This message announce the finish of the handshaking protocol from client side.

(iii) change cypher spec

→ This message is sent by server to show that it has made all the pending state of cypher protocol & parameter to active state.

finished of

→ This message announced the handshaking between server and finally is totally completed.

2. change cypher spec protocol

→ It is the simplest protocol. This protocol consists of single ~~bit~~ byte with value 0, 1.

→ It consist of single message only

→ It copies pending state to current state, which updates the cypher suite to be used to this connection.

3. SSL Alert Protocol

→ It is 2 byte protocol.

Level	Alert
Fatal warning	Error port
1 byte	1 byte

4. SSL record protocol

→ It is second sub-level protocol. It also called a lower level protocol. It provides following services —

- i) Encrypted data transmission
- ii) Encapsulation of data
- iii) Data Confidentiality
- iv) Data Integrity

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* SET protocol (Secure Electronic transaction)

→ SET is a security protocol designed to ensure the security & integrity of electronic credit & transaction.

It is developed with support from major organizations such as VISA, MasterCard, Microsoft (providing secure transaction technologies), & Netscape (SSL providing). SET employs advanced encryption & hashing technique for securing online credit card payments.

SET is not a payment system itself, it focuses on enhancing security in transactions.

Requirement & SET

- i) Mutual authentication :- Verify customer & merchant identity for intended uses.
- ii) Confidentiality :- Encrypt payment information (P.I) & order information (OI) security.
- iii) Message integrity :- Resist unauthorized modification to transmitted content.
- iv) Inter operability :- Ensure compatibility & seamless communication across system.

v) Optimal security mechanism :- Employed a best available security major for highest protection.

Participant in SET

- i) Card holder :- customer.
- ii) Issuer :- customer financial institution.
- iii) Certificate authority :- Authority that follow certain standard & issue certificate to all other participants.
- iv) Merchant.
- v) Acquirer :- Acquirer is merchant financial institution.